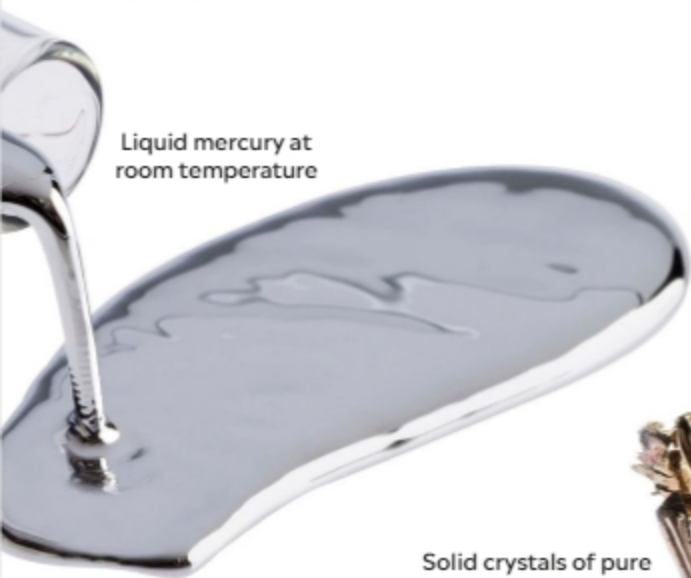


What is an element?

An element is a substance that cannot be broken down into simpler ingredients. Each one is made up of building blocks called atoms, which are unique for every element. For example, the element carbon contains only carbon atoms. There are 118 elements, and most, such as oxygen and gold, are found naturally on Earth—although rarely in their pure state. Elements are vital to our everyday lives, from the aluminum used to make kitchen foil to the zinc in our muscles and bones.



Liquid mercury at room temperature



Solid crystals of pure strontium refined in a laboratory



EYEWITNESS

Antoine Lavoisier

French chemist Antoine Lavoisier (right) helped in developing the science of chemistry in the 18th century. He tested scientific theories with careful experiments that featured accurate measurements. He also came up with an international system of naming chemical elements and compounds so that everyone could understand these names.



Elemental forms

All the elements exist in one of three primary states under normal conditions of temperature and pressure. They are either a solid, a liquid, or a gas, although a fourth special state called plasma is sometimes seen. At room temperature, most elements are solids. Only a few are gases, and only mercury and bromine are liquids.